

**Brand Visibility in AI Search: A Longitudinal Analysis
of AI Visibility Metrics in the U.S. Tea Industry**

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Abstract

No published study tracks competing brand AI visibility within a single consumer product category across generative AI platforms over time. This chapter addresses that gap through an analysis grounded in both the academic literature and sustained professional observation of AI search behavior in the U.S. tea industry. Using Whitebox, a generative engine intelligence platform, the study conducts longitudinal, observational analysis of AI visibility metrics for six major U.S. tea brands across ChatGPT and Google AI Overviews over ten weeks from November 2025 through January 2026. Two core metrics, mention rate and average position, were captured at five measurement intervals across 120 data points. The analysis documents five patterns: platform divergence, extreme volatility, mention-position decoupling, leadership instability, and category positioning effects. ChatGPT produced systematically higher mention rates (grand mean of 40.7%) than Google AI Overviews (22.3%), with cross-platform correlations by brand ranging from $r = -.445$ to $r = .820$. Of particular consequence, mention rate and average position operated independently within each platform, meaning that appearing more frequently did not produce more prominent placement. These patterns collectively demonstrate that AI visibility constitutes a competitive construct distinct from traditional search visibility and that effective brand representation in AI-generated responses depends on the same foundational investments in content quality and earned media that have always driven brand reputation.

Keywords: AI visibility, generative engine optimization, brand visibility, large language models, share of shelf, ChatGPT, Google AI Overviews, cross-platform measurement, consumer packaged goods, tea industry

Introduction

For more than two decades, search engine optimization governed how brands achieved digital visibility. The paradigm was stable in its essential structure: visibility was determined by position on a ranked list, optimization strategies were broadly transferable across platforms, and the relationship between ranking and click-through rate was well documented and predictable (Brin & Page, 1998; Dean, 2023). The introduction of ChatGPT in November 2022 and the rollout of Google AI Overviews in May 2024 disrupted this paradigm by replacing ranked lists with synthesized natural-language responses that select, organize, and present brand information through fundamentally different mechanisms. When a consumer asks a large language model for a tea recommendation, the system does not return a list of links. It constructs an answer, choosing which brands to include, in what order, and with what framing. The brand is either present in that answer or it is absent.

The commercial stakes are measurable. Adobe Analytics data indicate a 3,500 percent increase in U.S. retail site traffic from generative AI sources between July 2024 and May 2025 (Adobe, 2025). Pew Research Center (2025) documented that users who encountered an AI-generated summary clicked on a traditional search result link only 8 percent of the time, compared to 15 percent for users who did not encounter a summary. Bain & Company (2025) reported that approximately 80 percent of consumers now rely on zero-click results in at least 40 percent of their searches. A brand absent from the AI-generated response is not ranked lower. It is invisible.

The questions this study addresses did not originate in the literature. They originated in practice. Sustained work with a long-standing client in the U.S. tea industry, combined with direct observation of how ChatGPT, Claude, Gemini, and Perplexity responded to category and

brand queries, made clear that something structurally different was happening. The brands appearing prominently in AI-generated answers were not always the brands with the strongest search rankings or the largest share of shelf in traditional retail. The logic governing AI visibility was not the same logic governing search visibility and the gap between the two was consequential enough to warrant systematic investigation. This study began as a professional hypothesis. The data either confirm it or they do not. They do.

This chapter examines AI visibility dynamics for six major U.S. tea brands across ChatGPT and Google AI Overviews over a ten-week period spanning November 2025 through January 2026. Two metrics structure the analysis: mention rate, defined as the percentage of AI-generated responses in which a brand is referenced as a viable consumer option, and average position, the mean ordinal rank at which a brand appears within responses where it is mentioned. Three research questions organize the analysis: Do brands exhibit systematically different AI visibility profiles across ChatGPT and Google AI Overviews? Do mention rate and average position operate independently within each platform? And do AI visibility metrics exhibit substantial volatility across measurement intervals?

Background and Research Gap

The foundational work of Aggarwal et al. (2024) on Generative Engine Optimization established the conceptual distinction between traditional and generative search, demonstrating that optimization strategies effective in traditional SEO showed limited effectiveness in generative engine contexts while strategies involving citation addition and statistical enrichment produced significant visibility improvements. Subsequent experimental research demonstrated that large language model product recommendations are susceptible to manipulation through strategic content modifications (Kumar & Lakkaraju, 2024) and that cognitive bias language

systematically influences LLM recommendation behavior across multiple model architectures (Filandrianos et al., 2025). Most recently, Bagga et al. (2025) introduced E-GEO, the first benchmark specifically designed for e-commerce GEO, noting that current practices remain based on rules of thumb rather than systematically tested methods.

Platform theory provides the theoretical basis for expecting cross-platform divergence. The work of Cennamo and Santalo (2013) on differentiated platform ecosystems predicts that platforms operating under different architectural choices will produce different competitive outcomes for identical participants. ChatGPT and Google AI Overviews differ in training data, retrieval mechanisms, model architecture, and commercial incentives. Google AI Overviews is more tightly connected to web retrieval and structured authority signals. ChatGPT relies more heavily on learned associations within its training corpus and narrative coherence in response generation. These architectural differences generate the conditions for the platform divergence this study documents.

The motivation for this study was partly practical. The existing experimental literature, while methodologically rigorous, was built on fictitious product catalogs and single-platform designs. That work established that AI recommendation behavior is susceptible to manipulation and varies by content type, which are important findings. But no published academic research had examined how AI visibility actually behaves in a real industry, with established brands, genuine competitive history, and the kind of accumulated consumer sentiment that only develops over years of market presence. The tea industry is not an arbitrary choice. It is a category with distinct market segments, well-defined brand positioning across those segments, and a long enough history that the brands in this study carry real reputational weight in the sources AI platforms draw on. What the experimental literature could not answer, and what practitioners

needed to know, was whether the patterns observed with fictitious products hold when the brands are real, the competition is real, and the stakes are real. That question required a different kind of study.

Study Design

This study is a longitudinal, observational analysis of AI visibility metrics across two generative AI platforms for six major tea brands found in the United States: Bigelow Tea, Republic of Tea, Twinings, Harney and Sons, Traditional Medicinals, and Celestial Seasonings. These brands were selected to represent a range of market positions spanning mass-market, premium, and wellness-oriented segments. Data collection was conducted using the Whitebox generative engine intelligence platform across ChatGPT and Google AI Overviews at five measurement intervals between November 9, 2025, and January 19, 2026, yielding 120 data points across six brands, two platforms, two metrics, and five dates.

Each prompt was administered to 50 independent user sessions with clean memory, meaning each AI interaction was conducted in a fresh session state with no access to previous prompts or conversational context. This design prevents carryover effects and ensures each response reflects the model's default behavior for a first-time user. Prompts were organized into three thematic categories reflecting authentic consumer search behavior: green tea, black tea, and herbal tea. Representative prompts included queries such as "Looking for a green tea that isn't bitter, what's the best-tasting one for daily drinking?" and "I'm a heavy coffee drinker looking to switch to tea, any recommendations for a bold black tea with a kick?" Over 10,000 individual AI responses were collected per measurement interval, totaling more than 50,000 responses across the observation period.

Mention rate was calculated as the percentage of responses in which the brand was referenced as a genuine consumer recommendation. Average position was the mean ordinal rank at which a brand appeared within responses where it was mentioned. The analysis employs descriptive statistics and comparative methods appropriate to the exploratory nature of the study. No inferential statistical tests are employed, as the small number of measurement intervals precludes meaningful hypothesis testing in the traditional sense.

Findings

The analysis documents five empirical patterns: platform divergence, extreme volatility, mention-position decoupling, leadership instability, and category positioning effects.

Platform Divergence

ChatGPT produced systematically higher mention rates than Google AI Overviews across all five measurement intervals and all six brands. The grand mean mention rate on ChatGPT was .407 (SD = .155), compared to .223 (SD = .135) on Google AI, a difference of 18.4 percentage points. This gap was not driven by a single brand or a single date. It is a structural feature of the two platforms.

The divergence extends beyond aggregate levels to the rank ordering of brands. On ChatGPT, Republic of Tea ($M = .560$) and Bigelow Tea ($M = .540$) led in mention frequency. On Google AI Overviews, Twinings led ($M = .386$), followed by Bigelow Tea ($M = .313$). Cross-platform correlations by brand revealed substantial heterogeneity, ranging from $r = -.445$ for Traditional Medicinals to $r = .820$ for Republic of Tea. The same brand can lead on one platform and fall to the middle of the pack on the other.

The magnitude of this divergence was striking. Anyone accustomed to traditional search has a reasonable expectation of rough consistency across platforms: a user entering the same keyword into Google and Bing will encounter broadly similar results, because both systems are drawing on overlapping authority signals and link-based ranking logic. That expectation does not survive contact with the data presented here. The same consumer query directed at ChatGPT and Google AI Overviews can produce materially different brand hierarchies, with brands leading on one platform while falling to the middle of the pack on the other. Assuming consistency across AI platforms is not a conservative default. It is an error with measurable competitive consequences and it is an error that practitioners who have not yet examined cross-platform data are very likely making right now.

Extreme Volatility

The mean coefficient of variation for mention rate was 22.2% on ChatGPT and 33.9% on Google AI Overviews. Individual brands experienced single-interval swings exceeding 30 percentage points, with the largest change being Twinings on ChatGPT gaining 39.6 percentage points between December 20 and January 3. Google AI exhibits approximately 50% more volatility than ChatGPT on both metrics. Despite this instability, a durable competitive hierarchy persists beneath the surface: Kendall's W rank concordance across the five dates is .785 on ChatGPT and .743 on Google AI for mention rate, indicating moderate to strong concordance in the overall ordering of brands even as specific values shift substantially.

The volatility documented here does not mean that every fluctuation demands a response. Stock investors learn early that a single day's price movement tells you almost nothing useful about a stock's actual trajectory. Good portfolio management comes in knowing which changes are signal and which are noise, which only emerges over time. The same logic applies to AI

visibility data. A brand that sees its mention rate fall 15 percentage points between two intervals has not necessarily lost long-term ground if the broader pattern across several measurements remains stable or upward. Volatility findings do not require constant tactical adjustment but do warrant sustained observation for emerging directional patterns. Cross-industry patterns observed in the Whitebox dataset reinforce this approach: the brands that manage AI visibility most effectively are those that track consistently enough to distinguish a genuine shift in platform behavior from the ordinary turbulence of a non-deterministic system.

Mention-Position Decoupling

The study documents nine instances across both platforms where brands simultaneously gained in mention frequency while losing in positional prominence. The most pronounced case on ChatGPT is Traditional Medicinals between December 20 and January 3: mention rate rose from .258 to .481, an increase of 22.3 percentage points, while average position degraded from 3.4 to 5.9. The brand was included in nearly twice as many responses yet appeared in a less prominent position when included.

Across the full observation period, a different dimension of this decoupling emerged. Of the five patterns documented here, the mention-position decoupling was the most unexpected to encounter in the data. The intuitive assumption, carried over from two decades of search engine logic, is that a brand appearing more frequently would also appear more prominently. The data show the opposite can be true. Traditional Medicinals appeared in fewer responses than its competitors yet ranked in more prominent positions when it did appear, a pattern that only makes sense once you accept that inclusion and prominence are governed by separate algorithmic decisions. One is a relevance judgment. The other is a confidence judgment. They draw on overlapping but distinct signals and optimizing for one does not move the other.

Leadership Instability and Category Positioning Effects

No brand maintained the highest mention rate on both platforms simultaneously at any measurement point where the leaders diverged. On ChatGPT, the top brand changed three times across five intervals, with three different brands holding the lead. At two of the five measurement points, the platforms had different leaders entirely. The one exception to general instability is Twinings, which held the best average position on ChatGPT at all five measurement points, the only instance of sustained single-metric leadership in the dataset.

Traditional Medicinals, the only brand with an explicit wellness and specialty positioning, exhibited a distinctive pattern on Google AI Overviews. Its mean average position on Google AI was 1.92, compared to a mean of 3.14 for the remaining five brands. The brand appears in fewer responses but when it does appear, it tends to appear in a more prominent position. Supplementary prompt-level analysis shows Traditional Medicinals appears in approximately 35% of herbal and wellness-oriented prompts but only 15.8% of green tea prompts, confirming the positional advantage is driven by semantic fit between brand positioning and query intent rather than a general algorithmic preference.

Discussion

The five patterns documented here converge on a single conclusion: AI visibility constitutes a competitive construct distinct from both traditional search visibility and retail shelf visibility. In retail, share of shelf is a relatively stable metric governed by contracts, trade spending, and planogram decisions. In traditional search, a brand's share of the first page of results for category-relevant queries is deterministic, auditable, and changes incrementally. The data presented in this study demonstrate that AI visibility conforms to neither model. Share of AI

mention is neither stable, deterministic, nor governed by a single set of identifiable inputs. It varies dramatically across platforms, fluctuates by 20 to 40 percentage points within weeks, and operates on logic that separates inclusion from prominence.

The platform divergence finding has a direct implication the literature has not fully articulated. In the traditional search paradigm, optimization strategies developed for Google were broadly transferable to other search engines because ranking algorithms shared a common underlying logic rooted in link-based authority signals. The present data demonstrate this platform convergence does not hold in the generative AI environment. The divergence does not arise because the two platforms encounter different brand content. A brand produces one blog post, one FAQ page, one product description and both platforms can access it. The divergence arises because the platforms apply different evaluation logic to the same underlying content ecosystem. The content strategy can remain unified but the monitoring and the expectations must be platform-specific.

The mention-position decoupling extends platform theory in a direction the existing literature has not explored. In traditional two-sided markets, inclusion on the platform and prominence within the platform are typically governed by the same set of factors. The present data document a novel structural feature of generative AI platforms in which these two dimensions of visibility are operationally independent. AI systems first determine which brands are relevant to the query and belong in the response, and then separately determine the order in which those brands appear based on which ones seem more trustworthy or better suited to the specific context. For marketers accustomed to optimizing a single metric, this creates a fundamental question: whether to prioritize frequency of inclusion or prominence when included, knowing that gains in one do not reliably produce gains in the other. The answer is that both

require deliberate attention and progress on one cannot be assumed to produce progress on the other.

The category positioning effect observed for Traditional Medicinals suggests that brands with narrow, well-defined positioning occupy a structural advantage for queries that fall within their semantic territory. Brands that clearly stand for something specific are easier for AI systems to match to user queries. The advantage is not absolute, as Traditional Medicinals' low mention rate indicates it appears in fewer total responses. But when it does appear, it appears prominently, suggesting that Google AI Overviews assigns higher confidence to its relevance within specific query contexts.

Implications for Practice

Most brands are not yet tracking AI visibility as a defined metric and those that are aware of it are more likely to be grappling with the basic question of how to appear in AI-generated answers at all than with the distinction between mention frequency and positional prominence documented in this study. The dual-metric framework developed here is analytically useful and the data support it but it is ahead of where most marketing practitioners currently operate. Confusion about what AI visibility is and what to do about it is widespread and that confusion is not a failure of attention. The field itself is unsettled. Recommendations therefore need to meet practitioners where they are rather than assume a monitoring infrastructure that does not yet exist in most organizations.

The most counterproductive response to that uncertainty, and one visible in practitioner discussions across professional networks, is abandoning search engine optimization in favor of AI visibility investment. Negating a strategy that has produced measurable returns for years in order to pursue one whose mechanisms are still poorly understood is not a reasonable trade at

this stage of the field's development. The more defensible position is that the two are not in competition. The content quality, earned media relationships, and authoritative sourcing that support strong search performance are the same inputs that AI platforms draw on when constructing responses. A brand that has built genuine reputation does not need to choose. A brand that has not built it will find that neither search nor AI visibility forgives that absence for long.

Three operational priorities follow from the findings. The volatility documented here cannot be adequately monitored through quarterly reviews alone. Organizations that treat AI visibility as a periodic audit item risk missing directional shifts before they become consequential. The second priority is that monitoring and strategy must be separated by platform: the same brand performs differently on ChatGPT and Google AI Overviews and the two cannot be treated as interchangeable measures of the same construct. The third priority follows from the mention-position decoupling: a single optimization target is insufficient. Strategies that increase mention frequency may have no effect on positional prominence and the reverse is equally true. Both dimensions require deliberate attention.

For consumer packaged goods brands specifically, the practical actions translate to identifying which third-party sources AI platforms cite when responding to queries in the relevant product category, building partnerships with affiliate sites that appear in AI-generated responses, improving on-site pages that platforms are already sampling, and developing content in areas where competitors are being cited but the brand is not yet present. These are indirect strategies by necessity. The platform dependence dynamics described above mean that brands can identify which sources a platform draws on but cannot control how those sources are

weighted or predict how changes in their content ecosystem will translate into visibility outcomes.

Limitations and Future Research

Several constraints bound the claims this study can support. The ten-week observation window captures a temporal snapshot during a period of rapid platform development, and the specific magnitudes reported may not persist as the platforms mature. The five measurement intervals were not uniformly spaced, having been selected randomly for the tea industry tracking, which means the rate of change between consecutive intervals cannot be directly compared. The dataset of 120 data points precludes inferential statistical testing and the study's three hypotheses are evaluated against observed patterns rather than tested against null distributions.

The reliance on a single product category limits generalizability. While cross-industry data from the Whitebox platform suggest the documented patterns are not unique to tea, systematic confirmation requires replication across additional categories with distinct competitive structures and consumer search behaviors. The category positioning observation rests on a single brand and cannot establish a category-level effect without additional wellness and specialty brands in the sample. Geographic and linguistic boundaries also apply: because AI platforms adapt response generation based on IP location and query language, the visibility patterns documented here reflect U.S.-based, English-language query conditions specifically.

Future research should employ fixed-interval designs with uniformly spaced measurement points to enable cleaner temporal analysis. Cross-category replication, expanded platform comparisons to include Perplexity and emerging AI search tools, and causal investigation of the mechanisms through which content ecosystems shape AI visibility outcomes

would each advance the field. Longitudinal studies tracking brands through deliberate content interventions would begin to address the causal questions this observational study cannot answer.

Conclusion

The five patterns documented in this study, platform divergence, extreme volatility, mention-position decoupling, leadership instability, and category positioning effects, collectively demonstrate that AI visibility operates under dynamics structurally distinct from those governing traditional search engine visibility and retail shelf visibility. Brand representation in generative AI responses is neither stable nor deterministic. The metrics governing inclusion operate independently of those governing prominence and competitive position varies across platforms on a timescale that renders periodic optimization ineffective.

What the data ultimately demand is not a new discipline but a return to one that predates search engine optimization entirely. AI visibility is a metric worth tracking precisely because it reflects what consumers encounter when they turn to AI systems to learn about a brand, a category, or a purchase decision. But the underlying requirement is not algorithmic. Brands that produce genuinely expert, accurate, and useful content, and that do so consistently enough to earn coverage from sources AI platforms treat as authoritative, will find that AI visibility follows. This is what effective brand building has always required. The difference is that AI systems have removed the margin for neglecting it. Brands that have invested in content quality and earned media presence are discovering that investment now pays a second dividend. Those that have not are discovering they have less time than they thought to start.

Disclosure

Valorie Luther provides consulting services to Bigelow Tea through Creative Concepts. Bigelow Tea had no role in the design, data collection, analysis, or interpretation of this study and did not fund or review the research. Ofri Touboul-Cohen contributed as co-author in his capacity as Co-Founder and CEO of Whitebox which served as the data collection platform. Valorie Luther did not receive financial compensation from Whitebox.

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